

## JHARSUGUDA, India

WMO: 428860

Lat: 21.914N

Lon: 84.050E

Elev: 751

StdP: 14.30

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 12            | 51.7       | 53.9 | 33.0                          | 28.3 | 88.8 | 36.4 | 32.4 | 86.5 | 5.3                   | 75.6 | 4.6 | 76.0 | 0.1                      | 0    | 0.310 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |       |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|-------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%    |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB    | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)   | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 23.1                   | 109.2           | 72.3 | 107.0 | 72.3 | 104.4 | 72.9 | 81.8                | 90.1 | 81.0 | 89.0 | 80.6 | 88.3 | 2.9                  | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 80.0                            | 160.0 | 84.9 | 79.2 | 156.1 | 84.3 | 78.8 | 153.9 | 83.9 | 46.5          | 90.5 | 45.7 | 89.6 | 45.1 | 88.2 | 90.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 7.7           | 6.8  | 5.5 | 47.1                       | 112.6 | 1.9                | 1.6 | 45.7                                               | 113.7 | 44.6       | 114.6 | 43.6       | 115.5 | 42.2       | 116.7 |    |    |
| (5)               |      |     | 44.5                       | 85.4  | 1.9                | 1.9 | 43.2                                               | 86.7  | 42.1       | 87.8  | 41.1       | 88.9  | 39.7       | 90.2  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 80.9        | 68.6  | 74.6  | 82.6  | 90.3  | 93.2  | 89.1  | 82.8  | 82.6  | 83.0  | 80.5  | 74.3  | 68.9  |
|                                                                        | DBStd       | 8.49        | 3.51  | 4.28  | 4.36  | 4.30  | 3.78  | 5.72  | 2.67  | 2.19  | 2.07  | 2.93  | 3.35  | 3.36  |
|                                                                        | HDD50       | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 15          | 9     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 6     |
|                                                                        | CDD50       | 11272       | 575   | 689   | 1011  | 1209  | 1340  | 1172  | 1016  | 1012  | 989   | 944   | 729   | 586   |
|                                                                        | CDD65       | 5813        | 119   | 270   | 546   | 759   | 875   | 722   | 551   | 547   | 539   | 479   | 279   | 127   |
|                                                                        | CDH74       | 72786       | 1120  | 2819  | 6990  | 11813 | 14146 | 10314 | 6052  | 5870  | 5810  | 4611  | 2200  | 1041  |
|                                                                        | CDH80       | 37866       | 225   | 1255  | 4031  | 7833  | 9760  | 6179  | 2081  | 1861  | 2039  | 1742  | 682   | 178   |
| (14) Wind                                                              | WSAvg       | 1.3         | 0.6   | 0.9   | 1.1   | 1.7   | 1.9   | 2.3   | 2.2   | 1.8   | 1.3   | 0.8   | 0.4   | 0.4   |
| (15) Precipitation                                                     | PrecAvg     | 60.6        | 0.6   | 0.7   | 0.6   | 0.9   | 1.5   | 8.5   | 19.0  | 15.4  | 10.2  | 2.7   | 0.7   | 0.4   |
|                                                                        | PrecMax     | 95.4        | 4.0   | 2.0   | 2.1   | 3.6   | 7.2   | 20.9  | 35.7  | 25.7  | 26.2  | 8.4   | 3.7   | 4.2   |
|                                                                        | PrecMin     | 36.1        | 0.0   | 0.2   | 0.1   | 0.0   | 0.0   | 2.9   | 10.4  | 7.9   | 4.4   | 0.3   | 0.1   | 0.0   |
|                                                                        | PrecStd     | 11.5        | 0.9   | 0.4   | 0.5   | 0.9   | 1.4   | 4.0   | 5.9   | 4.9   | 4.7   | 2.1   | 0.9   | 0.9   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 88.0  | 96.3  | 104.2 | 110.5 | 112.1 | 110.5 | 94.6  | 92.3  | 93.3  | 93.3  | 89.5  | 86.1  |
|                                                                        |             | MCWB        | 64.5  | 65.3  | 66.6  | 70.2  | 72.8  | 74.9  | 80.3  | 80.3  | 79.0  | 75.1  | 70.5  | 65.5  |
|                                                                        | 2%          | DB          | 84.8  | 92.8  | 100.8 | 108.0 | 109.9 | 107.6 | 92.2  | 90.6  | 91.5  | 91.3  | 87.6  | 83.8  |
|                                                                        |             | MCWB        | 63.3  | 65.2  | 67.5  | 69.6  | 72.7  | 75.4  | 80.2  | 80.3  | 79.1  | 75.4  | 69.4  | 64.7  |
|                                                                        | 5%          | DB          | 82.5  | 90.4  | 98.5  | 105.9 | 107.6 | 103.6 | 90.2  | 89.1  | 90.1  | 89.6  | 86.0  | 82.0  |
|                                                                        |             | MCWB        | 62.5  | 64.6  | 67.4  | 69.8  | 73.2  | 76.9  | 79.9  | 80.0  | 79.2  | 75.4  | 68.7  | 63.9  |
|                                                                        | 10%         | DB          | 80.4  | 87.7  | 96.0  | 103.5 | 105.1 | 99.5  | 88.3  | 87.5  | 88.6  | 87.9  | 84.3  | 80.0  |
|                                                                        |             | MCWB        | 61.8  | 64.6  | 67.0  | 70.2  | 74.0  | 77.7  | 79.6  | 79.8  | 79.2  | 74.9  | 67.9  | 62.8  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 69.8  | 71.9  | 75.3  | 79.1  | 81.9  | 82.9  | 82.4  | 82.2  | 82.2  | 80.7  | 76.2  | 70.6  |
|                                                                        |             | MCDB        | 77.9  | 84.7  | 92.3  | 97.2  | 96.5  | 94.7  | 89.5  | 88.6  | 89.1  | 87.6  | 84.5  | 80.9  |
|                                                                        | 2%          | WB          | 67.4  | 69.9  | 73.4  | 76.8  | 80.5  | 81.4  | 81.2  | 81.4  | 81.2  | 79.5  | 74.3  | 68.3  |
|                                                                        |             | MCDB        | 76.9  | 82.6  | 90.6  | 93.8  | 96.2  | 92.8  | 87.9  | 87.5  | 87.9  | 86.4  | 81.8  | 76.8  |
|                                                                        | 5%          | WB          | 65.9  | 68.3  | 71.9  | 75.4  | 79.3  | 80.6  | 80.7  | 80.8  | 80.5  | 78.5  | 72.9  | 66.7  |
|                                                                        |             | MCDB        | 77.2  | 81.2  | 88.0  | 94.3  | 95.4  | 91.1  | 87.2  | 86.6  | 87.0  | 85.2  | 80.5  | 76.3  |
|                                                                        | 10%         | WB          | 64.4  | 67.0  | 70.3  | 74.3  | 78.1  | 79.9  | 80.1  | 80.1  | 79.9  | 77.5  | 71.4  | 65.2  |
|                                                                        |             | MCDB        | 76.2  | 80.6  | 86.6  | 93.7  | 94.5  | 90.3  | 86.1  | 85.5  | 85.9  | 83.8  | 79.2  | 75.7  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 26.0  | 26.2  | 26.3  | 25.4  | 23.1  | 16.1  | 9.7   | 9.4   | 11.2  | 16.5  | 22.6  | 25.0  |
|                                                                        |             | MCDBR       | 27.8  | 29.0  | 29.0  | 27.5  | 26.0  | 22.1  | 13.3  | 11.6  | 13.8  | 18.2  | 23.2  | 26.0  |
|                                                                        | 5% WB       | MCWBR       | 9.8   | 7.8   | 7.4   | 7.5   | 7.2   | 5.2   | 4.3   | 4.1   | 4.5   | 5.8   | 7.3   | 9.0   |
|                                                                        |             | MCWBR       | 22.3  | 22.5  | 23.9  | 24.3  | 22.5  | 16.3  | 11.4  | 10.6  | 11.8  | 13.9  | 16.7  | 21.3  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.469 | 0.473 | 0.537 | 0.556 | 0.616 | 0.671 | 0.592 | 0.567 | 0.568 | 0.541 | 0.518 | 0.480 |
|                                                                        |             | taud        | 2.016 | 2.041 | 1.897 | 1.895 | 1.799 | 1.719 | 1.905 | 1.988 | 1.971 | 1.987 | 1.946 | 2.006 |
|                                                                        | Ebn at Noon |             | 249   | 259   | 247   | 245   | 229   | 214   | 232   | 239   | 236   | 236   | 233   | 240   |
|                                                                        |             | Edn at Noon | 50    | 52    | 63    | 64    | 70    | 75    | 62    | 58    | 58    | 55    | 54    | 49    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1387  | 1657  | 1900  | 2122  | 2114  | 1642  | 1242  | 1264  | 1398  | 1525  | 1407  | 1302  |
|                                                                        | RadStd      |             | 85    | 90    | 87    | 93    | 58    | 166   | 151   | 90    | 119   | 137   | 79    | 79    |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page